

PRATHAM VANANGADU KESHAVA BABU

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Robotics Software Engineer | Embedded Linux | ROS 2 | Real-Time Control Systems

SUMMARY

Robotics Software Engineer with hands-on expertise in ROS 2, embedded Linux, real-time control, sensor fusion, and autonomous navigation. Built and deployed production robotics pipelines spanning YOLOv8 perception, Kalman/EKF state estimation, and A* motion planning from mathematical foundations to benchmarked, hardware-validated systems. UR-certified. Strong C++ and Python. Experienced with Gazebo, MoveIt 2, SLAM, NVIDIA Jetson, and CI/CD-backed embedded workflows.

TECHNICAL SKILLS

Backend & Frameworks: Spring Framework, Junit, Mockito
Databases: PostgreSQL, MongoDB, Redis, Azure SQL, AWS S3, Cosmos DB, Oracle, Hibernate.
Robotics & Middleware: Solidworks, ROS 2, MoveIt 2, Gazebo, SCADA, PLC, Isaac SIM, Nav2, SLAM, Path Planning (A*, RRT*, Dijkstra), Occupancy Grids, Universal Robots (UR-Certified)
Perception & State Estimation: YOLOv8, OpenCV, ByteTrack, LiDAR/RGB Fusion, Point Cloud Processing, Kalman Filter (1D/ND), EKF, IMU+GPS Fusion, Visual Servoing
Embedded & Systems: Embedded Linux, Embedded C++, Real-Time Systems, Yocto/Buildroot, NVIDIA Jetson, TFLite, Firmware Dev, OTA, Secure Boot
AI/LLM: Hugging Face, PyTorch, RAG, Neural Networks, Keras, Prompt Engineering, Lang Chain, N8N.
Languages & Tools: C++, Python (NumPy, OpenCV, FastAPI, Ultralytics), MATLAB (Simulink), Rust, AWS, Docker, Kubernetes, GitHub Actions, Azure DevOps, Git

WORK EXPERIENCE

Robotics Software Engineer | Internpro.ai, Tempe AZ Aug 2025 – Present

- Developed and deployed ROS 2 robotic systems on physical hardware with real-time control loops and Gazebo/MoveIt 2 sim-to-hardware pipelines improving object localization accuracy by 27%.
- Built vision-guided manipulation systems combining RGB/LiDAR sensor fusion, point cloud processing, and IK for autonomous pick-and-place boosting process efficiency by 70%.
- Engineered real-time visual servoing pipelines in low-light environments with closed-loop control, reducing task execution time by 95% on safety-critical operations.
- Designed dual-arm coordination and collision-aware motion planning with MoveIt 2; automated CI/CD (GitHub Actions) for embedded robotic software build/test/deploy.

Software Engineer | Ness Digital Engineering, Bengaluru Feb 2023 – Aug 2023

- Containerized Python FastAPI services with Docker; implemented JWT/Azure AD auth and CI/CD pipelines achieving 85%+ test coverage and reducing load times by 35%.
- Designed and developed microservices architecture in Java using Spring Boot, J2EE, JSP and JDBC within an Agile/SCRUM environment improving system scalability and reducing deployment time by 30%.
- Engineered asynchronous, fault tolerant distributed pipelines using Docker, Kubernetes, and Azure (CosmosDB and Data Factory) and AWS services (S3, EC2). Also wrote unit and Integration tests using JUNIT and Mockito, also optimized MongoDB query performance using Compass and Mongoose.

Machine Learning Intern | 1stop.ai, India Aug 2021 – Oct 2021

- Built FastAPI inference-serving endpoints for AI models on Linux with request validation and automated testing, build data-processing pipelines using Docker and Azure SQL, and integrated inference services into TypeScript web applications, enabling real-time predictions for internal users.
- Debugged API services, tuned endpoint performance and integration testing with Postman and Docker to achieve low latency responses.

PROJECTS

YOLOv8 Real-Time Robotics Perception & Safety System · Python, YOLOv8, ByteTrack, OpenCV, Supervision

- Built a production-grade AMR perception pipeline with YOLOv8 + ByteTrack multi-object tracking, configurable polygon danger zones, and a boolean E-stop safety signal — architecture mirrors ISO 3691-4 compliant warehouse robot safety systems.
- Benchmarked across 320×240→1280×720 resolutions; GPU inference >60 FPS on Jetson-class hardware. Frame-level JSONL telemetry logger enables offline incident replay and safety-event root-cause analysis.

A* Path Planning Engine with Algorithm Benchmarking · Python, NumPy, Matplotlib

- Zero-dependency 2D occupancy grid library implementing A*, Dijkstra, and Greedy best-first with a unified PlanResult interface matching Nav2 global planner foundations. A* explored 61% fewer nodes than Dijkstra on identical maps while guaranteeing shortest-path optimality.
- Interactive Matplotlib visualizer with animated exploration replay, live obstacle injection, and GIF export for clear algorithm comparison.

Kalman Filter Suite for Robot State Estimation (1D · 2D · EKF) · Python, NumPy

- Implemented 1D KF, N-D KF, and EKF for differential-drive robots [x, y, θ] with range/bearing landmark observations; Jacobian-linearised non-linear motion model directly applicable to GPS/IMU fusion matching robot_localization and Nav2 EKF node architecture in ROS 2.
- Validated quantitatively: 68% lower MAE vs raw sensor on 1D simulation; GPS scatter reduced from 3.0 m to 0.8 m on 2D curved trajectory. Dashboard includes Kalman gain convergence plots and 3σ uncertainty ellipses.

Digital Twin – UR5e Robotic Cell (Tic-Tac-Toe) · MATLAB/Simulink, SolidWorks, YOLOv5, Docker, SLAM

- SolidWorks digital twin of UR5e with MATLAB/Simulink autonomous control, SLAM localization, YOLOv5 game-state perception, and Docker CI/CD full perception→planning→control pipeline targeting physical UR hardware.

Embedded Real-Time Drone Orientation Detector · Embedded C++, TFLite, IMU, GitHub Actions

- On-device TFLite inference on microcontroller for IMU-based roll/pitch/yaw classification at 98.04% accuracy within real-time stabilization loop constraints; optimized for flash memory and latency via model compression and C++ firmware tuning.

Autonomous Robot Sim-to-Real Deployment Pipeline - ROS 2, Gazebo, NVIDIA Isaac Sim, YOLOv8, Git

- Engineered a multi-fidelity simulation architecture using Gazebo for lightweight, headless CI/CD regression testing and NVIDIA Isaac Sim Replicator for photorealistic synthetic data generation, quantifying the sim-to-real gap via real-world YOLOv8 mAP benchmarking.
- Optimized state estimation ruggedness by implementing a robot_localization EKF and writing automated scripts to map RMSE performance against progressive IMU noise injection, establishing predictable operational boundaries prior to physical deployment.

EDUCATION

MS, Systems Engineering – Robotics & Autonomous Systems | Arizona State University Tempe AZ, USA
Coursework: Robotics I, Mechatronics, Control Systems, Embedded ML, Real-Time Systems, Computer Vision. Aug 2023 – May 2025

B.E., Information and Computer Science | Visvesvaraya Technological University Bangalore, INDIA
Coursework: Operating Systems, System Design, Data Visualization, Linux/Unix, Neural Networks, LLD. Jul 2018 – Jun 2022